Keywords: omni-modal benchmarking • multimodal LLMs • ambiguity resolution • audio-visual integration

TRIAD: Benchmarking Omni-Modal Ambiguity in Multimodal Large Language Models

A New Benchmark Reveals That Even the Best Omni-Modal Systems Trail Humans by Over 20 Points When All Three Modalities Are Needed to Resolve a Scene

By Zhaolu Kang, Yidi Wang, Yile Li, Siqi Zeng, Yingjie He, Richeng Xuan, Zhichao Hu

arXiv:2610.30841 • NeurIPS 2026 Datasets & Benchmarks Track

Multimodal language models that process text, images, and audio are routinely evaluated on benchmarks where a single dominant modality carries the answer—leaving intact an unmeasured gap between claimed omni-modal capability and the genuine ability to fuse all three modalities simultaneously. TRIAD closes this gap with 327 hand-crafted queries spanning 86 scene groups and 18 ambiguity categories, each designed so that the complete text–image–audio triplet yields a unique answer while any single-modality subset leaves two or more options open. Across 14 state-of-the-art omni-modal systems, the best model trails humans by over 20 group-accuracy points, and removing answer options drives model group accuracy to single digits.

AT A GLANCE

Problem: Existing benchmarks do not require models to integrate all three modalities; a single modality usually suffices to answer correctly.

Why it matters: Omni-modal systems are deployed as universal assistants; a model that silently ignores audio will fail in safety-critical settings where acoustic context is essential.

Method: 327 text–image–audio triplets where the correct answer requires all three modalities, evaluated on item and scene- group accuracy with and without answer options.

Result: Best model trails humans by >20 points on group accuracy; removing answer options drops model group accuracy to single digits.

Evidence: 14 state-of-the-art omni-modal systems on 86 scene groups covering 18 ambiguity categories.

The Big Picture

The ambition of omni-modal language models—processing text, image, and audio within a single system—promises agents that perceive the world as richly as humans do. Yet evaluating this ambition has relied on benchmarks that are functionally bimodal: image-question answering where text alone is sufficient, or audio tasks where the image adds marginal signal. True omni-modal integration requires all three modalities to be jointly necessary.

TRIAD operationalizes this requirement through the principle of *modal indispensability*: for each query, every modality is necessary and no proper subset is sufficient. The dataset captures everyday scenes where text, image, and audio together resolve what neither pair can—e.g., hearing an alarm while seeing a clock while reading a date.

Why Existing Approaches Fall Short

Item-level evaluation is misleading. A model that achieves 70% accuracy on individual items may still fail every scene group, because group accuracy requires consistent disambiguation across all items sharing a scene. TRIAD reports both metrics; the gap between them exposes unreliable partial reasoning.

Answer options inflate performance. Multiple-choice formats let models exploit statistical regularities in the option set (e.g., always picking the longest option). TRIAD evaluates both with and without answer options; performance drops to single digits in the open-ended condition, confirming option exploitation.

Audio cues are underweighted. Models trained on predominantly text–vision data default to text–image reasoning even when the disambiguating signal is acoustic. TRIAD's audio-indispensable categories expose this systematic underweighting.

Core Mechanism

TRIAD's benchmark construction follows three stages:

1. **Scene collection:** Authors collect or synthesize naturalistic scenes representable in all three modalities with complementary but partially overlapping information.
2. **Ambiguity verification:** Each triplet is manually reviewed to confirm that any two-modality combination leaves at least two consistent answer options.
3. **Query authoring:** Questions are written to be answerable only with the full triplet; distractors are crafted to be plausible given incomplete modality access.

Scene groups (86 in total) share a common real-world context and each contain 3–5 queries, allowing the group accuracy metric to test whether disambiguation general-

izes within a scenario.

Technical Formulation

Let $\mathcal{T} = (t, v, a)$ denote a triplet of text t, image v, and audio a, and let $\mathcal{A} = \{a_1, \ldots, a_K\}$ be the answer set. A triplet is modally indispensable if:

$$|\arg\max_k P(a_k \mid \mathcal{T})| = 1 \quad \text{but} \quad |\arg\max_k P(a_k \mid \mathcal{T}\backslash\{m\})| > 1$$

for every modality $m \in \{t, v, a\}$. This condition defines the 327 items in TRIAD; it is verified by human annotators, not automatically.

Group accuracy for scene group g is:

$$\text{GroupAcc}(g) = \mathbf{1}[\forall q \in g,\ \hat{a}(q) = a^*(q)]$$

where $\hat{a}(q)$ is the model's predicted answer and $a^*(q)$ the ground-truth. Mean group accuracy across all 86 groups is the primary metric.

Learning or Inference Procedure

TRIAD is a static evaluation benchmark; no training is involved. Models are prompted with the triplet in their native input format (text + image + audio), asked to output either a multiple-choice letter or a free-form answer, and scored automatically. Human performance is estimated from a held-out annotator pool distinct from the construction team.

What the Guarantee Says

The modal-indispensability design guarantee is:

- **Soundness:** A model that answers correctly must have accessed all three modalities (no single-modality shortcut exists by construction).
- **Completeness:** Group accuracy measures consistent disambiguation within a scene, ruling out lucky individual item guesses.

These are guarantees about the benchmark's diagnostic validity, not about any particular model's capability.

Experimental Findings

- **Group accuracy gap:** The best of 14 evaluated omni-modal systems achieves group accuracy >20 points below human performance.
- **Item vs. group accuracy:** Models with high item accuracy (e.g., >65%) still fail most scene groups, confirming inconsistency.
- **Open-ended collapse:** Removing answer options drops average model group accuracy to single digits; human group accuracy declines only modestly (from ∼85% to ∼70%).
- **Audio underuse:** On audio-indispensable items, models score near chance even in the multiple-choice condition, confirming that acoustic reasoning is the primary failure mode.

Ablations and Interpretation

Modality ablation: Providing models with only text+image, text+audio, or image+audio gives performance close to random, confirming benchmark validity.

Model family comparison: Closed-source frontier models outperform open-source models by a larger margin on TRIAD than on existing benchmarks, suggesting that scale and instruction tuning improve omni-modal integration but do not close the gap.

Category breakdown: The hardest ambiguity categories involve temporal audio events (e.g., distinguishing two events from their sounds alone) and spatial sound-source localization, pointing toward specific architectural improvements.

Reference

Zhaolu Kang, Yidi Wang, Yile Li, Siqi Zeng, Yingjie He, Richeng Xuan, Zhichao Hu. **TRIAD: Benchmarking Omni-Modal Ambiguity in Multimodal Large Language Models**. arXiv:2610.30841. NeurIPS 2026. https://arxiv.org/abs/2610.30841

Keywords: multi-objective reinforcement learning • ensemble policies • trajectory attribution • objective conflict

Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution

Objective-Conflict Coefficients Quantify Preference Trade-offs, While Routed Ensembles and Kernel-Surrogate Attribution Close the Multi-Objective Efficiency Gap

By Zhenshuo Zhang, Ruoxuan Xiong, Hongyang R. Zhang

arXiv:2610.35733 • NeurIPS 2026

Multi-objective reinforcement learning demands a single agent to simultaneously satisfy multiple reward objectives weighted by a user-supplied preference vector—a task made hard when objectives conflict and gradient directions oppose each other. This paper introduces objective-conflict coefficients that formally quantify this conflict, derives matching online regret bounds, and proposes two practical algorithms: a routed ensemble that routes each trajectory to the basis policy with highest expected return under its preference weight, and a trajectory attribution module based on kernel ridge regression that identifies which trajectories are most valuable for which preference objectives.

AT A GLANCE

Problem: A single preference-conditioned policy fails when objectives strongly conflict; the degree of conflict was not previously formalised.

Why it matters: MORL is central to robotics, resource allocation, and fairness-aware learning, where trade-offs among objectives must be navigated at deployment.

Method: Objective-conflict coefficients + routed ensemble policy learning (trajectory routing to K basis policies) + kernel ridge regression trajectory attribution.

Result: Consistent improvements over single-policy and standard ensemble baselines on multi-objective robotic control benchmarks.

Evidence: Multi-objective MuJoCo locomotion benchmarks; formal regret analysis as a function of maximum conflict coefficient.

The Big Picture

In multi-objective reinforcement learning (MORL), the agent receives a vector reward $\mathbf{r} = (r_1, \ldots, r_K)$ and a preference weight $\mathbf{w} \in \Delta^{K-1}$. The goal is to find a policy π that maximises $\mathbf{w}^\top J(\pi)$ across many possible $\mathbf{w}$, effectively learning the entire Pareto front of preference-weighted returns.

Conflict arises when $\nabla_\theta J_i(\pi_\theta)$ and $\nabla_\theta J_j(\pi_\theta)$ point in opposing directions, forcing any gradient step to damage at least one objective. Prior approaches either concatenate $\mathbf{w}$ to the state (ignoring conflict) or use hand-crafted ensemble partitions (ignoring data efficiency). This work addresses both gaps.

Why Existing Approaches Fall Short

Conditioned single policy: Appending $\mathbf{w}$ to the state does not prevent gradient conflicts. For strongly conflicting objectives, the single policy converges to a suboptimal compromise across all preferences rather than achieving the Pareto front.

Naïve ensembles: Manually partitioning the preference space into K fixed regions assigns trajectories to basis policies by region membership, ignoring that trajectories generated under one preference weight may be highly informative for another.

No formal measure of conflict: Without a quantitative measure of objective conflict, there is no principled way to predict how many basis policies are needed or which trajectory assignments are most beneficial.

Core Mechanism

Routed Ensemble Policy Learning maintains K basis policies $\{\pi_1, \ldots, \pi_K\}$ and iterates:

1. Collect trajectories $\{\tau_j\}$ by sampling basis policies according to the current preference weight distribution.
2. **Route:** Assign each τ_j to the basis policy with the highest expected return under τ_j's preference weight: $k^*(\tau_j) = \arg\max_k \mathbf{w}_j^\top J_k(\pi_k)$.
3. **Update:** Apply PPO to each π_k using only trajectories routed to it.

Trajectory Attribution uses a kernel ridge regression surrogate $\hat{f}(\pi, \mathbf{w}; \mathcal{D})$ to estimate policy performance as a function of the training trajectory set $\mathcal{D}$. Leave-one-out attribution scores ϕ_j^k identify which trajectories help or hurt each basis policy under each preference weight, enabling curriculum sampling.

Technical Formulation

Objective-Conflict Coefficient. For objectives i and j:

$$c_{ij}(\pi_\theta) = \frac{-\nabla_\theta J_i^\top \nabla_\theta J_j}{\|\nabla_\theta J_i\| \, \|\nabla_\theta J_j\|}$$

A positive c_{ij} indicates conflict (gradients oppose); the maximum over all pairs and all preference weights bounds the MORL regret:

$$\text{Regret}(T) = O\Big(\sqrt{T \cdot c_{\max}}\Big)$$

where $c_{\max} = \max_{i \neq j, \mathbf{w}} c_{ij}(\pi^*(\mathbf{w}))$.

Kernel Surrogate. Let $\phi(\tau, \mathbf{w})$ be a feature embedding of trajectory τ and preference weight $\mathbf{w}$. The surrogate is:

$$\hat{f}(\mathbf{w}; \mathcal{D}) = \mathbf{y}^\top (K + \lambda I)^{-1} k(\mathbf{w})$$

where $K_{ij} = \kappa(\phi(\tau_i), \phi(\tau_j))$, $\mathbf{y}_j = \mathbf{w}_j^\top J(\pi)$, and λ is a regularisation constant. The attribution of trajectory τ_j for preference $\mathbf{w}$ is the leave-one-out change in $\hat{f}$.

Learning or Inference Procedure

Each training step:

1. Sample a batch of preference weights $\{\mathbf{w}_j\}_{j=1}^B$ from the prior distribution over Δ^{K-1}.
2. Collect trajectories by rolling out each π_k under its currently assigned preference weights.
3. Compute attribution scores; upsample high-attribution trajectories.
4. Route trajectories to basis policies and apply PPO updates.

At test time, given a user preference $\mathbf{w}^*$, the system evaluates $J_k(\pi_k)$ for each basis policy and returns the best.

What the Guarantee Says

The regret bound $O(\sqrt{T \cdot c_{\max}})$ provides an instance-dependent characterisation: MORL is hard when objectives conflict strongly ($c_{\max}$ large) and easy when they are compatible ($c_{\max}$ small). The routed ensemble achieves this rate for any $K \geq c_{\max}/\epsilon$ for tolerance ϵ. This is the first formal characterisation of MORL hardness in terms of objective geometry.

Experimental Findings

- **Multi-objective MuJoCo locomotion:** Routed ensemble outperforms conditioned single-policy baselines by 8–15% in hypervolume metric across 4 multi-objective task suites.
- **Versus naïve ensemble:** Attribution-based curriculum sampling improves sample efficiency by ∼30% (same hypervolume with fewer environment interactions).
- **Conflict coefficient:** Empirical $c_{\max}$ correlates strongly with the observed performance gap between single-policy and ensemble methods ($r = 0.81$), validating the theoretical predictor.
- **Scalability:** Performance improvements hold for $K \in \{2, 4, 8\}$ basis policies; gains plateau at $K = 8$.

Ablations and Interpretation

Routing vs. random assignment: Random assignment of trajectories to basis policies reduces hypervolume by 7%, confirming that conflict-aware routing is the key driver.

Attribution vs. uniform sampling: Replacing kernel attribution with uniform trajectory sampling costs ∼20% in sample efficiency, isolating the contribution of the surrogate.

Conflict coefficient threshold: Task suites with $c_{\max} < 0.1$ show no significant gain from ensembles; the single-policy baseline already matches. This provides a practical criterion for when ensembles are needed.

Reference

Zhenshuo Zhang, Ruoxuan Xiong, Hongyang R. Zhang. **Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution**. arXiv:2610.35733. NeurIPS 2026. https://arxiv.org/abs/2610.35733

Keywords: video diffusion • novel view synthesis • autonomous driving simulation • geometry grounding

GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation

LiDAR-Warped Reference Frames and Synthetic Para4D Data Enable Zero-Shot Lane-Shift Video Generation on Real-World Waymo and nuScenes Benchmarks

By Yuchen Xi, Yuhan Guo, Chenwei Hou, Zixu Liu, Jin Fang, Jason Liu, Ruigang Yang

arXiv:2610.25245 • ECCV 2026

Autonomous driving simulators require realistic video from camera poses that were never recorded—a lane shift, a slightly different heading, or a hypothetical trailing vehicle. GeoV2V addresses this challenge by combining LiDAR and depth-based geometric priors with full-video conditioning on the Wan 2.1 video diffusion backbone, generating temporally consistent, geometrically accurate driving videos under large trajectory shifts. Trained entirely on Para4D, a new synthetic CARLA-based dataset of synchronized multi-lane videos, GeoV2V transfers zero-shot to real-world Waymo and nuScenes data, achieving state-of-the-art visual quality and geometric consistency.

AT A GLANCE

Problem: Large viewpoint shifts in novel-view driving video reveal occluded regions; pure generation loses geometric coherence across frames.

Why it matters: High-fidelity novel-view synthesis is essential for scalable simulation of rare events (lane changes, near-misses) that cannot be captured by fixed sensor rigs.

Method: Geometric warping of reference frames via LiDAR/depth priors + Wan 2.1 video diffusion conditioned on warped sequence + Para4D synthetic training data.

Result: State-of-the-art FID, FVD, and depth-consistency on Waymo and nuScenes without fine-tuning on real data.

Evidence: Waymo, nuScenes, and Para4D benchmarks; ablations isolating geometric conditioning and synthetic pretraining.

The Big Picture

The long tail of driving scenarios—rare lane changes, unexpected obstacles, adverse weather—cannot be covered by fixed-route data collection. Novel view synthesis (NVS) addresses this by generating plausible videos from camera poses never recorded. For autonomous driving, NVS must satisfy two requirements that pure appearance-based generation fails to meet: (1) geometric consistency across time, and (2) faithfulness to 3D scene structure rather than statistical plausibility.

GeoV2V grounds a powerful video diffusion prior (Wan 2.1) with explicit 3D geometry, turning the generation problem into a geometry-aware inpainting problem: warp what is visible, inpaint what is occluded, and preserve temporal dynamics.

Why Existing Approaches Fall Short

Image-based NVS methods (NeRF, 3DGS): These methods require dense multi-view captures or long sequences for 3D reconstruction—unavailable in drive-by single-camera settings.

Pure video generation (Sora, Gen-3): Unconstrained video generation ignores 3D structure, producing temporally inconsistent lane layouts and incorrect depth ordering.

Warping-only methods: Geometric warp alone leaves holes where occluded content is revealed by the viewpoint shift; filling them with inpainting lacks temporal coherence across the full video.

Core Mechanism

GeoV2V operates in three stages per generated video:

1. **Geometric warping:** Given the reference video and a target viewpoint trajectory, LiDAR point clouds (or monocular depth where LiDAR is unavailable) project each reference frame to the target pose, producing a warped reference sequence with explicit holes where geometry is occluded.
2. **Conditional video generation:** The Wan 2.1 video diffusion transformer takes the warped reference sequence as a conditioning signal via cross-attention, filling holes and refining geometric artifacts while preserving large-scale 3D structure.
3. **Para4D training:** The model is trained on Para4D, a synthetic multi-lane CARLA dataset providing pairs of synchronized cameras at different lateral offsets under diverse weather, lighting, and traffic conditions—the only source of explicit cross-lane supervision.

Technical Formulation

Let $V_{\text{ref}} = \{I_t^{\text{ref}}\}_{t=1}^{T}$ be the reference video and P_{tgt} the target camera pose sequence. The warp operator $\mathcal{W}$ uses depth D_t and camera intrinsics K to project pixels:

$$\tilde{I}_t^{\text{tgt}} = \mathcal{W}(I_t^{\text{ref}}, D_t, P_t^{\text{ref}} \to P_t^{\text{tgt}}, K)$$

yielding a warped sequence $\tilde{V} = \{\tilde{I}_t^{\text{tgt}}\}$ with per-pixel validity masks M_t marking visible regions.

The video diffusion model G_θ generates the output:

$$V_{\text{out}} = G_\theta\big(z_T, \tilde{V}, M\big)$$

where z_T is Gaussian noise and conditioning on $(\tilde{V}, M)$ is implemented via cross-attention at each transformer layer. The training objective is the standard flow-matching loss $\mathcal{L} = \mathbb{E}[\|G_\theta(z_t, \tilde{V}, M) - V_{\text{gt}}\|^2]$ on Para4D pairs.

Learning or Inference Procedure

Training: GeoV2V is trained entirely on synthetic Para4D data (no real-world data). The model sees diverse CARLA scenes with lateral offsets of 1.0–5.0 m, covering standard lane shifts and extreme viewpoints. Wan 2.1 weights are initialised from a pretrained checkpoint and fine-tuned with the geometric conditioning added.

Inference (zero-shot transfer): On Waymo and nuScenes, only LiDAR-based depth is used for warping; the generator runs without any domain-specific fine-tuning, relying on the synthetic-to-real generalisation of the learned geometric prior.

What the Guarantee Says

The geometric warping step provides an explicit constraint: visible pixels in the reference frame are reproduced at the correct target location with sub-pixel registration accuracy (verified by LiDAR reprojection error). The diffusion model only hallucinates content in provably occluded regions (mask $M_t = 0$), giving a formal separation between geometric reliability and generative uncertainty.

Experimental Findings

- **Waymo lane-shift NVS:** State-of-the-art FID and FVD against prior methods (WoVoGen, GeoFlow, AnyScene), with >15% improvement in FVD at 3.5 m lane shifts.
- **nuScenes:** Consistent gains in geometric depth consistency (homography error reduction >20%) and visual quality metrics.
- **Challenging lighting:** GeoV2V maintains FVD gains in night and adverse weather conditions where geometric priors are noisiest.
- **Para4D ablation:** Replacing Para4D with a warping-only baseline increases FVD by 38%, confirming the importance of explicit cross-lane supervision.

Ablations and Interpretation

LiDAR vs. monocular depth: LiDAR warping improves depth consistency by 12% over monocular depth; monocular depth is sufficient for visual quality (FVD) but not for geometric accuracy (reprojection error).

Conditioning mechanism: Replacing cross-attention conditioning with concatenation reduces FVD by 8%, confirming that the temporal context of the warped sequence is best exploited through attention.

Training data scale: Para4D with 10k scene pairs is sufficient; doubling to 20k improves FVD by only 2%, suggesting diminishing returns from synthetic data.

Reference

Yuchen Xi, Yuhan Guo, Chenwei Hou, Zixu Liu, Jin Fang, Jason Liu, Ruigang Yang. **GeoV2V: Geometry-Grounded Video Diffusion Model for Driving Scene Generation**. arXiv:2610.25245. ECCV 2026. https://arxiv.org/abs/2610.25245

Keywords: pixel diffusion transformers • prediction parameterization • MFlow adaptive interpolation • ConvRefiner

Rethinking the Design Space of Vanilla Pixel Diffusion Transformers

MFlow Learns a Per-Direction Interpolation Between x_0- and Velocity Prediction, Closing the Quality Gap Between Pixel and Latent Diffusion

By Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra

arXiv:2610.23854 • Under review, 2026

Pixel-space diffusion transformers avoid VAE tokenization artifacts but have lagged behind latent diffusion in generation quality, partly because the optimal prediction parameterization—clean image (x_0) versus velocity (u)—depends non-trivially on model capacity and data statistics. MFlow addresses this by introducing per-principal- direction gains that interpolate between x_0- and u-prediction, learned end-to-end with diffusion training. Experiments on ImageNet, UCF-101 video, and large-scale text-to-image generation confirm that MFlow consistently outperforms fixed-parameterization baselines and that larger models systematically prefer velocity prediction. A lightweight ConvRefiner trained post-hoc with an adversarial loss further closes the persistent high-frequency gap.

AT A GLANCE

Problem: The best prediction parameterization for pixel DiTs varies with model capacity and data; the fixed-choice approach leaves quality on the table.

Why it matters: Pixel DiTs promise simpler, higher-fidelity generation than latent diffusion, but their quality gap has blocked adoption.

Method: Per-direction gain $m_i \in [0,1]$ interpolates x_0- and u-prediction; gains are trained jointly with the diffusion model. ConvRefiner handles residual high-frequency deficits post-hoc.

Result: Competitive or improved FID/FVD vs. fixed-parameterization baselines on ImageNet, UCF-101, and text-to-image.

Evidence: Ablations across model scales, architectures, and modalities; systematic m_i analysis validating the capacity- preference relationship.

The Big Picture

Latent diffusion models (LDMs) such as SD3 and FLUX compress images into a low-dimensional latent code before diffusion, enabling fast generation but introducing VAE reconstruction artifacts. Pixel diffusion transformers (pixel DiTs) sidestep this by diffusing directly in pixel space, preserving sharp boundaries and avoiding compression bottlenecks.

The quality gap between pixel DiTs and LDMs has been attributed to less stable training and slower convergence. This paper identifies an underappreciated cause: the prediction parameterization. The choice between predicting the clean image (x_0), the noise (ε), or the velocity ($u = \alpha x_0 - \sigma\varepsilon$) determines the gradient signal across noise levels and model capacity in ways that have not been studied systematically for pixel DiTs.

Why Existing Approaches Fall Short

Fixed x_0-prediction (JiT): JiT showed that x_0-prediction outperforms ε-prediction for pixel DiTs at moderate scale, but this advantage diminishes or reverses at larger model sizes. The fixed choice becomes suboptimal as models scale.

Fixed u-prediction (flow matching): Velocity (flow matching) prediction works well for latent diffusion but underperforms x_0-prediction at smaller pixel DiT scales, because low-capacity models lack the expressiveness to accurately predict the velocity at all noise levels.

AsymFlow: A binary rank-asymmetric assignment of x_0 vs. u prediction to different principal components is a special case of MFlow with step-function gains, and loses the benefit of smooth interpolation.

Core Mechanism

MFlow parameterizes the prediction target for the i-th principal direction of the data as:

$$\hat{y}_i = m_i \cdot u_i + (1 - m_i) \cdot (x_0)_i$$

where the scalar gain $m_i \in [0,1]$ is the sole additional parameter per direction. Setting $m_i = 0$ recovers clean-image prediction; $m_i = 1$ recovers velocity prediction. Gains are initialized from empirical signal-to-noise statistics and updated with gradient descent as part of the standard diffusion training loop.

ConvRefiner is a lightweight CNN trained to predict the residual between a generated pixel DiT image and the real image in the frequency domain. It is trained post-hoc with a spectral adversarial loss while keeping the diffusion backbone frozen, adding only modest compute.

Technical Formulation

Let $x_0 \in \mathbb{R}^d$ be a clean image in pixel space. The flow matching diffusion process defines a noisy image:

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$$

The velocity target is $u = \alpha_t x_0 - \sigma_t \varepsilon$. Under MFlow, the model f_θ predicts:

$$\hat{y}_i(x_t, t) = m_i f_\theta^{(u)}(x_t, t)_i + (1 - m_i) f_\theta^{(x_0)}(x_t, t)_i$$

The training loss is the standard flow-matching objective:

$$\mathcal{L}_{\text{MFlow}} = \mathbb{E}_{t, x_0, \varepsilon}\left[\sum_i \left\|\hat{y}_i(x_t, t) - y_i^{\text{target}}\right\|^2\right]$$

where $y_i^{\text{target}} = m_i u_i + (1 - m_i)(x_0)_i$. Gradients flow through both f_θ and $\{m_i\}$.

The ConvRefiner adds:

$$\hat{x}_{\text{refined}} = G_{\text{ref}}(\hat{x}_0) + \hat{x}_0$$

with G_{ref} trained to minimize a spectral GAN loss between $\hat{x}_{\text{refined}}$ and real images.

Learning or Inference Procedure

Training: Standard diffusion training with an extra gradient term for $\{m_i\}$ at each step. The gains are initialised as $m_i = \text{SNR}_i/(1 + \text{SNR}_i)$ where SNR_i is the signal-to-noise ratio along principal component i at the median noise level. ConvRefiner is trained in a second stage with the backbone frozen.

Inference: Identical to standard flow-matching sampling (DDIM, DPM-Solver, or Euler method). The gain m_i does not affect inference speed. ConvRefiner adds one forward pass with the lightweight CNN after the diffusion chain.

What the Guarantee Says

The MFlow loss is a strict generalization of fixed x_0- and u-prediction: setting all $m_i = 0$ recovers the x_0 loss, setting all $m_i = 1$ recovers the velocity loss. Therefore, MFlow can only improve on either fixed choice—its optima weakly dominate—assuming sufficient optimization.

For capacity scaling, the paper proves that for a linear model, the optimal m_i^* satisfies:

$$m_i^* = \frac{\text{Var}[u_i]}{\text{Var}[u_i] + \text{Var}[(x_0)_i]}$$

which is an increasing function of the intrinsic dimensionality of the i-th component—larger models, which can represent more directions, benefit more from velocity prediction.

Experimental Findings

- **ImageNet 256×256:** MFlow reduces FID by 15–20% over fixed x_0- and u-prediction baselines at the same model scale.
- **UCF-101 video:** Consistent FVD improvement across all model sizes; gains are largest for the largest model (+18% over velocity-only baseline).
- **Large-scale text-to-image:** MFlow at 1B parameters matches or exceeds the prior best pixel DiT at 3B parameters.
- **ConvRefiner:** Adds 1.2 ms per image and reduces high-frequency FID deficit by 40% without changing global FID.
- **Capacity scaling:** Empirical m_i values shift from near-zero to near-one as model size increases, confirming the theoretical prediction.

Ablations and Interpretation

Gain initialization: Random initialization of m_i converges to similar final values but requires 2× more training steps; SNR-based initialization is strongly recommended.

Per-direction vs. global gain: A single shared gain m for all directions reduces FID by only 60% of the per-direction improvement, confirming that different principal components benefit differently.

ConvRefiner training signal: Replacing the spectral GAN loss with an ℓ_2 loss in pixel space fails to recover high-frequency detail—the adversarial objective is essential.

Reference

Soumik Mukhopadhyay, Long Mai, Abhinav Shrivastava, Aniruddha Mahapatra. **Rethinking the Design Space of Vanilla Pixel Diffusion Transformers**. arXiv:2610.23854. Under review, 2026. https://arxiv.org/abs/2610.23854

Keywords: test-time adaptation • robot locomotion • reinforcement learning • distribution shift

TEMPO: Efficient Test-Time Adaptation For Robot Policies

Critic Entropy Regularization and Score-Based Memory Enable On-Device Policy Adaptation—Up to 80% Performance Recovery on Physical Quadrupeds Under Load

By Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos

arXiv:2610.20318 • NeurIPS 2026

Reinforcement-learned locomotion policies trained in simulation degrade in deployment when conditions shift—payload changes the center of mass, terrain modifies effective friction, sensors drift with temperature. TEMPO addresses this with a surprisingly effective combination: an entropy regularizer on the critic that prevents value collapse under distribution shift, paired with a score-based memory buffer that upweights hard observations during on-device adaptation. Across three quadruped platforms (Go1, Go2, G1) and four shift types in MuJoCo Playground, TEMPO yields up to 80% improvement in the single-device setting; on a physical Unitree Go2 carrying an added payload, TEMPO recovers pre-shift performance within minutes.

AT A GLANCE

Problem: Sim-to-real locomotion policies degrade when deployment conditions (payload, terrain, sensors) differ from training—and no simulator is available at test time.

Why it matters: Real-world deployment requires policies that adapt autonomously; simulator-based fine-tuning is impractical on hardware.

Method: Critic entropy regularizer prevents value collapse; score-based memory oversamples observations the policy finds hard.

Result: Up to 80% performance improvement in simulation; physical Unitree Go2 recovers from payload shift.

Evidence: Go1, Go2, G1 in MuJoCo Playground under four distribution shifts; physical Unitree Go2 experiment.

The Big Picture

Reinforcement learning in simulation remains the dominant approach for robot locomotion, but the sim-to-real gap means deployed policies face conditions their training never covered. The standard remedy—domain randomisation—widens the training distribution but cannot anticipate all deployment shifts. Once deployed, the robot's own experience is the only data available for further improvement.

Test-time adaptation (TTA) of neural networks has been studied for supervised classification, but RL policies present a harder challenge: there are no ground-truth labels, the reward is expensive to evaluate, the critic can diverge under distribution shift, and compute is constrained to a single embedded device. TEMPO is the first method to address all four constraints simultaneously.

Why Existing Approaches Fall Short

Standard PPO fine-tuning at test time: Without a simulator, PPO must use on-device rollouts directly. The off-policy nature of the limited experience combined with a shifting observation distribution causes the critic to diverge, destabilising policy updates.

Domain randomisation: Broad randomisation reduces peak sim performance and still fails for shifts outside the randomised range (e.g., a specific payload not anticipated during training).

Meta-learning (MAML, PEARL): Meta-learning requires a distribution over adaptation tasks at training time—unavailable when the real-world distribution is unknown—and inner-loop gradient computation that is expensive on embedded hardware.

Core Mechanism

TEMPO adds two components on top of PPO, requiring no changes to the policy network or training procedure:

1. Critic Entropy Regularizer. The critic loss is augmented with an entropy penalty:

$$\mathcal{L}_V = \mathcal{L}_{\text{TD}} + \beta\, H(V_\phi(s))$$

where $H(V_\phi(s))$ is the entropy of the value distribution across a replay batch. This prevents the critic from collapsing to a near-constant value function under distribution shift, which would zero out policy gradient signals.

2. Score-Based Memory. Each observation s in the replay buffer is assigned a score proportional to its TD error:

$$\text{score}(s) \propto |r + \gamma V_\phi(s') - V_\phi(s)|$$

Observations with high scores—those where the critic is most wrong—are sampled more frequently, biasing updates toward the hardest parts of the new distribution. Scores are updated online as the critic improves.

Technical Formulation

The full TEMPO objective at each adaptation step is:

$$\mathcal{L}_{\text{TEMPO}} = \mathcal{L}_{\text{PPO}} + \lambda_V \left[\mathcal{L}_{\text{TD}} + \beta\, H(V_\phi)\right]$$

where $\mathcal{L}_{\text{PPO}}$ is the clipped surrogate objective and λ_V balances the critic and actor losses.

The score-based sampler maintains a priority queue over N stored transitions, replacing the oldest with probability proportional to inverse score (keeping recent hard examples longer). Sampling probability for transition i is:

$$p_i = \frac{\text{score}(s_i)^\alpha}{\sum_j \text{score}(s_j)^\alpha}, \quad \alpha = 0.6$$

(Priority exponent α follows the PER convention.)

Learning or Inference Procedure

Offline training (simulation): Standard PPO with domain randomisation. No modifications to the training procedure.

Online adaptation (deployment): At each deployment step:

1. Execute the current policy on the robot; collect transitions.
2. Update transition scores from critic TD errors.
3. Sample a batch from the priority queue.
4. Apply one PPO + entropy-regularised critic update step.

Total compute per step: equivalent to one standard PPO update, feasible on the onboard CPU/GPU of a Unitree robot.

What the Guarantee Says

Under mild assumptions (bounded reward, Lipschitz value function, score ergodicity), the entropy regularizer prevents the critic from collapsing to a degenerate fixed point during distribution shift:

$$H(V_\phi(s)) \geq H_{\min} > 0 \;\text{ for all } t$$

ensuring policy gradient variance stays non-zero. This provides a necessary condition for adaptation progress; a convergence rate depends on the magnitude of the distribution shift and the data collection speed.

Experimental Findings

- **Go2 (MuJoCo Playground, payload shift):** TEMPO reaches 80% of pre-shift performance within 5 minutes of on-device interaction; standard PPO fine-tuning diverges.
- **Go1, G1 (MuJoCo Playground, four shift types):** Consistent gains across all platforms and shift types; average +42% over non-adapted baseline.
- **Physical Unitree Go2:** TEMPO with a 2 kg added payload recovers stable locomotion within 10 minutes; the non-adapted policy fails to sustain a trot gait.
- **Baseline comparison:** Outperforms domain randomisation, meta-RL (MAML), and test-time normalisation across all conditions.

Ablations and Interpretation

Entropy regularizer only: Without score-based memory, the critic stabilises but adaptation is slower (-25% performance at 5 min); both components are needed.

Score-based memory only: Without entropy regularization, the critic still diverges in 3 of 4 shift scenarios, confirming that stability must come before curriculum selection.

Priority exponent α: Performance is robust to $\alpha \in [0.4, 0.7]$; $\alpha < 0.4$ approaches uniform sampling and loses most gains.

Batch size: Adaptation improves up to batch size 64 and plateaus, consistent with the variance reduction argument underlying critic entropy regularization.

Reference

Motasem Alfarra, Pietro Mazzaglia, Markus Peschl, Daniel Dijkman, Christos Louizos. **Efficient Test-Time Adaptation For Robot Policies**. arXiv:2610.20318. NeurIPS 2026. https://arxiv.org/abs/2610.20318